Predicting Chicago 311 Service Requests After Winter Storms

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1. Introduction

The goal of this project is to design and implement a model that can predict the number of requests received by the City of Chicago's 311 City Services department for road maintenance and other city services following winter weather events. Chicago's climate often involves harsh winters that can bring large amounts of snow and ice and dangerously cold temperatures. These conditions often lead to roads and sidewalks being impassible, extensive damage to roadway surfaces due to the repeated use of snow plows and road salt for snow and ice removal, damage to the city's water utilities due to freezing, and residents needing a place to shelter in the event they lose indoor heating or plumbing. During a winter storm, City Services often experiences a spike in the number of requests they receive, and this increased demand puts a strain on city resources and delays response times. Our model will help city officials better plan for increased requests following a weather event and determine the areas most likely to be in need of service. The model will be trained based on City Services' ticket history combined with historical NOAA climate data and, if time allows, demographic information about each ward within the city. The model will use upcoming weather forecasts to estimate in advance the timing of an expected rise in the number of service requests, and provide a projection of how many requests can be expected for City Services, as well as an approximation for the wards from which the most requests will be received.

2. Research Goals

This model will be designed to evaluate the effects of winter weather events on Chicago's roadway and public utility infrastructure, and help determine whether upcoming weather forecasts and demographic data can be used to accurately project the number of service requests received from each ward to reliably allocate city resources and personnel for rapid response to such requests.

2.1 Research Questions

The research goal can be broken down into these specific questions:

1. Based on existing data, how does winter weather typically affect Chicago's roadway and utility infrastructure across each of its wards?
2. How does each winter weather event generally impact the number of service requests received by City Services, and what is the usual lag time between the weather event and the first signs of a spike in the number of tickets submitted?
3. Does the implementation of this prediction model significantly improve the accuracy of projected ticket numbers following a weather event?

3. Data Description

Service request records from 311 City Services will provide ticket data from across the city. Each row represents one request, and contains columns for specific information related to that request. The full dataset has 14.8 million rows and 39 columns, covering from 12/18/2018 to present. Relevant columns include the request type, the date created, the completion date (if applicable), the full street address of the location where service is needed (including street number, street name, and ZIP code) as well as the specific community area and ward the address is located in, and a flag variable indicating whether the request is a duplicate of an existing record.

Historical weather data will be obtained from the National Oceanic and Atmospheric Administration's climate data portal. Each row in the dataset represents a single day's weather conditions as recorded by a single weather station within the coverage area. Each column is one attribute of the daily record, including the specific weather station, the date, and various measurements for that day's conditions. The model will focus on daily snowfall, cumulative snow depth (amount of snow on the ground any given day), average daily wind speed, and daily low temperature.

3.1 Data Characteristics, Population, and Challenges

3.1.1 Target Population vs. Sample Observations

- Target Population: The ideal dataset would consist of all instances of service being needed by residents citywide as a direct result of winter weather. As for the climatic condition dataset, it should include the true weather conditions across all the wards at all times.
- Observed Sample: The original available sample consists of only the instances of service being needed that were actually reported to City Services via phone call to 311 or online ticket submission. The weather data sample includes the atmospheric conditions and climate details recorded specifically at the Chicago O'Hare Airport in the given timeline.

3.1.2 Expected Challenges

- Reporting Delay Bias: Because the reporting of the infrastructural damages relies on residents self-reporting them, in conditions of extreme weather like snowstorms or freezing temperatures, there might be delays in travelling or submitting complaints related to infrastructure, thus creating an anticipated 1- to 3-day reporting lag.
- Weather Data Granularity: The NOAA dataset consists of records from one location (O'Hare Airport) rather than from all the involved wards. This can introduce potential microclimate approximation gaps.
- Spatial and Demographic Disparities: Prior urban research indicates that the citizen reporting frequency varies across neighbourhoods depending on the level of community engagement they possess, meaning lower ticket count in certain areas may reflect reporting gaps rather than good road conditions.

- Dataset Cleaning Hurdles: The dataset needs to be cleaned and formatted such that no duplicate data records for the same physical incident exist. The empty cells also need to be explicitly zero filled or eliminated depending on its relevance.

4. Methodology Plan

4.1 Candidate Models

We plan to predict the number of service requests for each Chicago ward on each day. Because the target is a count (0, 1, 2, ...), we use two tree-based models and compare them against simple baselines:

(a) LightGBM (gradient boosted trees), our main model.
(b) Random Forest, a second model for comparison.

Why LightGBM:
- It supports a Poisson loss, which is designed for count data and does not predict negative values.
- Weather effects are not simple straight lines. For example, heavy snow followed by a thaw may cause more damage than either one alone. Tree-based models learn these combinations automatically.
- It works well with lag features (weather from 1 to 7 days ago), which lets us study the delay between a storm and a spike in requests.
- It trains quickly on our expected table size, so tuning settings and time-based cross-validation are practical.
- It shows which features matter most, which helps explain results to city officials.

Why Random Forest:
- It builds many trees independently and averages them, while LightGBM builds trees one after another, each fixing the mistakes of the previous ones. The two work in different ways, so comparing them tells us whether boosting is worth using on our data.
- It is less likely to overfit with little tuning, which matters because we expect only about 7 winters of data.
- It also handles combinations of weather conditions and supports a Poisson criterion for counts.

The two models are trained separately on the same data, and we compare their scores on the same test period. We keep the one with the lowest test error as our final model.

4.2 Training Strategy

Data setup: Each row is one ward on one day. The target (Y) is the number of requests that day. The inputs (X) are:
- Weather: snowfall, snow depth, temperature, and a freeze-thaw flag, from the NOAA O'Hare station.
- Lagged weather: the same values 1, 2, 3, 5 and 7 days ago.

- Forecast weather: expected snowfall and temperature 1 to 3 days ahead.
- Calendar: day of week, month, holidays.
- Recent requests: the number of requests in the same area over the previous 7 days.
- Ward information: area ID and, if time allows, census demographics.

Training steps:

1. Clean the 311 data and keep only the request types related to winter weather services, and filter out the summer months when winter weather does not occur.
2. Join the weather and ward data to the daily table.
3. Split the data by time, not randomly. We train on earlier winters, tune on a middle period, and test on the most recent winter. This prevents the models from seeing the future.
4. Tune each model's settings (for example tree depth and number of trees; learning rate for LightGBM) using rolling time-based cross-validation.
5. Train LightGBM and Random Forest separately, both with Poisson loss.
6. Evaluate both on the test winter using MAE, RMSE, and Poisson deviance. We also check if each model's top 10 predicted busiest areas match the real top 10 for each storm.
7. Report results separately for storm days and normal days, since most days are quiet.
8. Use feature importance (or SHAP) on the final model to explain which weather factors matter most.

Forecast weather: In real use, the city would only have weather forecasts, not actual weather. Because archived historical forecasts may be hard to get, we plan to use archived forecasts or actual weather with added random error to imitate forecast mistakes. We note this as a limitation, since results may look slightly better than in real use.

4.3 Baseline & Reasoning

Before testing whether weather helps, we need a benchmark for how well requests can be predicted without it. We use three simple history-based forecasts per ward: persistence (the latest daily count), seasonal naive (the same weekday one week before), and a 7-day moving average. We evaluated the baselines on road-surface damage requests from the Chicago 311 dataset (December 2018 onward), aggregated to daily counts for each of the 50 wards. The test set is winter 2025-26 (Dec 1 to Mar 31; 121 days; 6,050 ward-days; mean 4.1 requests per ward-day, about 207 citywide per day), held out as a whole season because requests are strongly seasonal. Forecasts are made 1, 2 and 3 days ahead using only data available at the forecast date. We report MAE, RMSE, and surge-day MAE (the busiest 10% of test days by citywide total, about 12 days), and we also sum ward forecasts to citywide totals so results can be compared at both levels.

Model	Horizon (days)	Ward MAE	Ward RMSE	Ward surge MAE	Citywide MAE
Persistence	1 / 2 / 3	3.99 / 4.43 / 4.48	7.25 / 7.86 / 7.91	7.17 / 7.56 / 7.37	85 / 127 / 132

Seasonal Naive	1 / 2 / 3	4.07 / 4.07 / 4.07	7.50 / 7.50 / 7.50	7.51 / 7.51 / 7.51	85 / 85 / 85
7-Day Moving Average	1 / 2 / 3	3.44 / 3.48 / 3.48	5.87 / 5.97 / 5.98	5.93 / 6.07 / 6.20	88 / 90 / 90

At ward level, the 7-day moving average is the strongest baseline (MAE about 3.5) and is the benchmark our models must beat, since ward counts are small and sparse (24% of ward-days are zero) and averaging a week of data beats copying a single past day. Summed to citywide, seasonal naive has the lowest MAE at 2 and 3 days ahead (85, roughly 41% of average daily volume) and is essentially tied with persistence at 1 day, because the weekly cycle dominates at larger volumes. All baselines make larger errors on surge days, roughly 1.7 to 1.9 times their overall ward error (ward surge MAE 5.9 to 7.5 vs. 3.4 to 4.5), since history alone cannot anticipate freeze-thaw or snowfall events, which is the gap our weather features target. Limitations: one test winter, so surge results (about 12 days) are indicative only; requests without a ward (0.9% of test-winter requests) were excluded; duplicates not removed; ward boundaries changed in 2023, which does not affect the baselines but must be handled when training models.

5. References

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